

WWE-UIE: A Wavelet & White Balance Efficient Network for Underwater Image Enhancement with Novel Architecture Extensions

Project Presentation

1. Problem Statement

Research Paper:

WWE-UIE: A Wavelet & White Balance Efficient Network for Underwater Image Enhancement (WACV 2026)

Problem

- Underwater images suffer from:
 - Color distortion due to wavelength-dependent light absorption
 - Low contrast and visibility
 - Blurring caused by light scattering
 - Loss of edge and texture details
- Existing Transformer-based and diffusion-based UIE methods achieve good performance but require:
 - Large model sizes • High FLOPs • Slow inference
- **Goal:** Design a lightweight UIE model that preserves quality while enabling real-time deployment — and propose novel components that further improve quality *without* increasing FLOPs meaningfully.

2. Experimental Setup

Datasets

- UIEB (890 images)
- UFO-120 (1620 images)
- LSUI (4279 images)
- EUVP-Scene (2185 images)
- EUVP-Dark (5500 images)

Implementation

- Framework: PyTorch
- Input: 256×256
- Optimizer: AdamW, $\text{lr} = 2 \times 10^{-3}$
- Epochs: 100(paper) 50(us),
Batch size: 24-4

Evaluation Metrics

- *Full-reference* :
 - PSNR, SSIM
- *Non-reference*:
 - UCIQE, NIQE
 - URanker, CIEDE2000

Our baseline run (UIEB, batch=4)

PSNR: 22.13 SSIM: 0.897
(close to paper; gap due to smaller batch due to computation power)

Proposed model (batch=4)

PSNR: **24.15** SSIM: **0.912**

3. Proposed Methodology

WWE-UIE Backbone (reproduced)

- 1 **White Balance**
fixes color imbalance by balancing the rgb values
- 2 **WaveletEnhanceBlock (WEB)**
Splits image into 4 parts based on frequencies of information and then learns them separately such as color textures and structures
- 3 **SGFB**
Makes edges more sharper after web block
- 4 **U-Net** encoder-decoder plain + skip connections

Our Novel Contributions

1. Wavelet Subband Attention (WSAM)

After Haar DWT the 4 subbands (LL/LH/HL/HH) are fused blindly. **WSAM** adds SE-style attention *per subband group*, letting the model suppress noise-heavy bands (HH) and amplify structure bands (LL) adaptively. ≈ 0 FLOPs overhead.

2. Adaptive Cross-Channel Correction (IACCC)

Replaces Gray-World heuristic with a tiny MLP that predicts a **3×3 color matrix per image** from its own RGB statistics. Models Beer-Lambert cross-channel attenuation. 316 parameters, < 1 MAC per image, FGSC

4. Comparison with State-of-the-Art Methods

Method	UIEB PSNR	SSIM	Params (M)
NU2Net	22.96	0.8840	3.15
UDAFFormer	23.50	0.9020	9.59
LiteEnhanceNet	22.39	0.8810	0.013
SMDR-IS	23.43	0.9010	12.25
PhaseFormer	23.76	0.9110	1.78
WWE-UIE (paper b=24)	24.32	0.919	0.734
WWE-UIE (paper b=4)	22.13	0.897	0.734
WWE-UIE + Ours b=4	24.15	0.912	0.736

Key Findings

- WWE-UIE already achieves highest PSNR with only 0.734M params
- Our additions (WSAM + IACCC + FGSC) add only ~ 0.002 M params
- PSNR almost as good as baseline with smaller batch size as well.
- All 3 components are drop-in replacements in model.py; no architecture redesign needed

5. Ablation Study and Conclusion

Original Paper Ablation (UIEB)

Configuration	PSNR	SSIM
w/o White Balance	24.18	0.9175
w/o WEB	24.22	0.9134
w/o SGFB Gradient	23.90	0.9143
Swapped Order	24.06	0.9154
Full WWE-UIE	24.32	0.9196

Conclusion

- Achieves strong enhancement quality with very low computational cost.
- Demonstrates an effective balance between:
 - Accuracy
 - Model size Inference speed

Conclusion

WWE-UIE provides a strong lightweight baseline combining:

- White Balance Prior
- Wavelet Enhancement
- Gradient-aware Refinement

We additionally propose:

- **WSAM** — subband-aware wavelet attention
- **IACCC** — cross-channel color correction
- **FGSC** — frequency-gated skip connections

Takeaway